

Machine Learning

1. Programming Fundamentals for ML

- **YouTube:**

- [Python for Data Science Full Course by freeCodeCamp](#) – Complete end-to-end breakdown of python execution logic, functions, and data structures.

- [Corey Schafer - Python OOP Series](#) – The ultimate industry baseline for writing clean, object-oriented, reusable production code.

- **Hands-on Labs:**

- [Exercism \(Python Track\)](#) – Algorithmic programming problems with automated tests and human-in-the-loop mentor reviews.

- [Kaggle Learn: Python](#) – Fast, interactive browser-based environments handling core syntax structures.

- **Documentation:**

- [NumPy User Guide](#) & [Vectorization Tutorials](#) – The definitive structural optimization manuals for matrix manipulation.

2. Data Collection, Sources, and Preprocessing

- **YouTube:**

- [CampusX Feature Engineering for Machine Learning](#) – Deep dive into imputation, structural encoding, outliers, and mathematical data scaling.

- [Keith Galli's Pandas Playlist](#) – High-yield, clear implementation guides sorting, filtering, and cleaning messy tabular data frames.

- **Hands-on Labs:**

- [Kaggle Learn: Data Cleaning](#) – In-browser sandbox environments focused on handling missing values, scaling, and character encodings.

- [UCI Machine Learning Repository](#) – Download raw, completely unformatted data matrices to practice building preprocessing pipelines manually.

- **Documentation:**

- [Scikit-Learn Preprocessing Guide](#) – Mathematical explanations behind standard scalers, target encoders, and vectorization methods.

3. Machine Learning Fundamentals and Learning...

- **YouTube:**

- [StatQuest with Josh Starmer - Machine Learning Foundations](#) – Unmatched visual animations that break down bias-variance trade-offs, gradient descent, and

cost functions simply.

- [Stanford CS229: Machine Learning by Andrew Ng](#) – The legendary, comprehensive lecture series covering foundational statistical learning theory.

- **Hands-on Labs:**

- [Google Developers Machine Learning Crash Course](#) – Interactive coding modules with real-time feedback loops built directly into your browser.

- [Scikit-Learn Example Gallery](#) – Ready-to-run python execution scripts simulating classical machine learning concepts.

- **Documentation:**

- [Introduction to Statistical Learning \(ISLR Textbook\)](#) – The ultimate theoretical blueprint book for understanding underlying model behaviors.

4. Supervised Learning Algorithms

- **YouTube:**

- [DataTalksClub Machine Learning Zoomcamp](#) – The definitive, software-engineering approach to building tree models, regression variants, and boosting.

- [StatQuest: Random Forests and Gradient Boosting \(XGBoost\)](#) – Detailed visual proofs explaining how weak predictive models combine into robust systems.

- **Hands-on Labs:**

- [Kaggle Competitions \(Titanic / House Prices\)](#) – The absolute industry proving ground for dialing in complex algorithmic configurations.

- [Scikit-Learn Classifier Sandboxes](#) – In-browser scripting environments for deploying and testing support vector machines, Naive Bayes, and linear models.

- **Documentation:**

- [XGBoost Documentation & LightGBM Core Guides](#) – Technical engineering manuals detailing hyperparameter optimization.

5. Unsupervised Learning and Reinforcement...

- **YouTube:**

- [Stanford CS229: Unsupervised Learning Lectures](#) – Academic, rigorous mathematical breakdowns of K-Means, hierarchical clustering, and PCA logic.

- [Hugging Face Deep Reinforcement Learning Course](#) – Exceptional step-by-step videos tracking Q-learning, policy gradients, and actor-critic models.

- **Hands-on Labs:**

- [Gymnasium \(Farama Foundation / Open AI Gym\)](#) – The global standard environment sandbox to train RL agents in physics engines, retro games, and text spaces.

- [Google Developers Clustering Course](#) – Guided practice modules modeling

structural data similarity matching and segmentation.

- **Documentation:**

- [Scikit-Learn Unsupervised Modules & Stable-Baselines3 Docs](#) – Complete programming manuals for dimensionality reduction and baseline RL agents.

6. Model Evaluation, Validation, and Metrics

- **YouTube:**

- [Valerio Velardo - The Sound of AI \(Model Evaluation\)](#) – Practical engineering perspective on cross-validation layouts, stratified splitting, and avoiding data leakage.

- [StatQuest: ROC, AUC, Bias, and Variance](#) – Visual guides mapping threshold trade-offs, confusion matrices, and performance pitfalls.

- **Hands-on Labs:**

- [Weights & Biases \(W&B\) Practical Quickstarts](#) – Live dashboard tracks for logging model metrics, validation loops, and training hyperparameter runs.

- [Neptune.ai Sandbox Playground](#) – Connect machine learning scripts directly to centralized, modern experiment trackers.

- **Documentation:**

- [Scikit-Learn Model Evaluation Metrics](#) – Highly structured directory detailing precision, recall, F1, Log Loss, and custom cost parameters.

7. Deep Learning Fundamentals

- **YouTube:**

- [PyTorch for Deep Learning Course by Daniel Bourke](#) – A completely hands-on, massive blueprint covering tensor operations, backward propagation, and dataset utilities.

- [DeepLearning.AI Deep Learning Specialization](#) – Andrew Ng's iconic series outlining neural network parameters, optimization techniques, and regularization.

- **Hands-on Labs:**

- [TensorFlow Playground](#) – A beautiful visual web environment showing neural networks tuning weights and active bias nodes in real-time.

- [Fast.ai: Practical Deep Learning for Coders](#) – Top-down engineering track dedicated to building and deploying neural patterns safely.

- **Documentation:**

- [PyTorch Tutorials & TensorFlow Core Guides](#) – The industry-standard reference manuals for constructing custom neural architecture layers.

8. CNNs, RNNs, and Transformer Architectures

- **YouTube:**

- Andrej Karpathy's Neural Networks: Zero to Hero – The finest educational series on the internet, showing you how to build tokenizers and code a GPT Transformer from scratch.
- Hugging Face NLP Course Videos – Seamless implementation guides for attention layers, self-attention mechanics, and language model fine-tuning.
- **Hands-on Labs:**
 - Hugging Face NLP Course Chapters – In-browser sandbox tracks for building computer vision encoders, tokenizers, and Transformer architectures.
 - PyTorch Image Models (timm) Quickstarts – Practice pulling, modifying, and fine-tuning state-of-the-art vision networks locally.
- **Documentation:**
 - Attention Is All You Need (Original Paper) & Hugging Face Docs – Foundational theory and code recipes for cutting-edge deep learning systems.

9. MLOps, Deployment, and Production ML

- **YouTube:**
 - DataTalksClub MLOps Zoomcamp – The undisputed gold standard tracking models from local python scripts up to scalable cloud infrastructure.
 - TechWorld with Nana MLOps Explained – High-level structural reviews of model registries, tracking engines, pipeline steps, and serving environments.
- **Hands-on Labs:**
 - MLflow Guided Integration Scenarios – Spin up local tracking infrastructure to manage model versioning, artifacts, and parameters.
 - Evidently AI Monitoring Tutorials – Code scripts modeled around finding data drift, concept drift, and performance decay over time.
- **Documentation:**
 - The MLOps Community Roadmap & Google Cloud Architecture Framework (MLOps) – Enterprise-grade deployment playbooks for continuous model delivery.